

Meeting 3

Lex Janssens

Date:	15 Apr 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
Mentor:	dr.ir. Á. Serra-Gomez
Project term:	February 2025 – August, 2025
E-mail:	l.p.janssens@umail.leidenuniv.nl

Outcome of previous meeting

- Find a more complex Mujoco environment to experiment on.
- Check which hyperparameters are overlapping and which are inbuilt → find out what could be the cause of the difference in performance.
- Run experiments with and without wrappers in Mujoco environment setting.
- Find out how to support Mujoco-v1. → Experiment to align results of original PPO paper with SB3 and CleanRL using the same hyperparameters to expose consistency between frameworks and original algorithm.

Results during the last weeks

- Singled out Mujoco environment, which is more complex but in time complexity manageable.
- Fixed some bugs within my code and added an evaluation function as a common metric.
- Compared the two PPO implementations of SB3 and CleanRL, in terms of (inbuilt) hyperparameters and methodologies. Found two differences that may result in difference in performance.
- Ran SB3 and CleanRL PPO on HalfCheetah-v4, with and without wrappers provided by CleanRL, including observation normalization and transformation, and reward normalization and transformation.
- Still stuck on getting Mujoco-py to work.

Plans for next weeks

- Find out if the implementation of GAE in CleanRL is actually different.
- A last shot at Mujoco-v1 (since I found another (older) binary repository).
- TBD